

EDA DATA CLEANING

AIM

To perform Exploratory Data Analysis and Data Cleaning on the California Housing Dataset by examining the dataset structure, dimensions, data types, statistical characteristics, missing values, duplicate records, housing-value distributions, ocean-proximity-wise housing patterns, housing-feature-wise statistics, correlation between numerical attributes, top housing records based on house value and income, and exporting the cleaned dataset for further analysis.

ALGORITHM

1. Import the required libraries such as **Pandas and NumPy**.
2. Load the **California Housing Dataset** into a Pandas DataFrame.
3. Display the **first five and last five records** of the dataset.
4. Display the **dataset information, shape, column names, and data types**.
5. Generate **descriptive statistical information** for numerical attributes.
6. Identify the **missing values** present in each column.
7. Remove **completely blank rows and duplicate records** from the dataset.
8. Standardize the **ocean proximity** categorical attribute by removing unwanted spaces and converting values to uppercase.
9. Convert numerical attributes into appropriate **numeric data types** and handle missing numerical values using **median imputation**.
10. Analyze the **statistical characteristics of median house values**.
11. Analyze the distribution of **house values across different ocean proximity categories**.
12. Analyze important housing features such as **total rooms, total bedrooms, population, households, median income, and median house value**.
13. Calculate the **correlation between numerical attributes** in the dataset.
14. Identify the **top 10 housing records based on median house value**.
15. Identify the **top 10 records based on median income**.
16. Export the cleaned dataset into **CSV, Excel, and JSON formats**.
17. Display a confirmation message after successful creation of the cleaned dataset files.

CODE

```
import pandas as pd
import numpy as np

# Load dataset
path = "/content/housing prices.csv"
```

```
df = pd.read_csv(path)
```

```
print("\nFirst 5 rows:")
```

```
display(df.head())
```

```
print("\nLast 5 rows:")
```

```
display(df.tail())
```

```
print("\nDataset information:")
```

```
df.info()
```

```
print("\nShape:")
```

```
print(df.shape)
```

```
print("\nColumns:")
```

```
print(df.columns.tolist())
```

```
print("\nData types:")
```

```
print(df.dtypes)
```

```
print("\nStatistical description:")
```

```
display(df.describe())
```

```
print("\nMissing values:")
```

```
display(df.isnull().sum())
```

```
# Data cleaning
```

```
df = df.dropna(how="all")
```

```
df = df.drop_duplicates()
```

```

df["ocean_proximity"] = (
    df["ocean_proximity"].astype(str).str.strip().str.upper()
)

numeric_columns = df.select_dtypes(include=np.number).columns

for col in numeric_columns:
    df[col] = pd.to_numeric(df[col], errors="coerce")
    df[col] = df[col].fillna(df[col].median())

print("\nMissing values after cleaning:")
display(df.isnull().sum())

print("\nCleaned dataset shape:")
print(df.shape)

# Housing value analysis
print("\nHousing Value Statistics:")
display(
    df["median_house_value"].agg(
        ["count", "mean", "median", "std", "min", "max"]
    )
)

# Ocean proximity analysis
print("\nOcean-Proximity-wise Housing Analysis:")
display(
    df.groupby("ocean_proximity")["median_house_value"]
        .agg(["count", "mean"])
        .sort_values("mean", ascending=False)

```

```
)
```

```
# Feature-wise analysis
```

```
print("\nHousing Feature-wise Statistics:")
```

```
display(
```

```
    df[
```

```
        [
```

```
            "total_rooms",
```

```
            "total_bedrooms",
```

```
            "population",
```

```
            "households",
```

```
            "median_income",
```

```
            "median_house_value"
```

```
        ]
```

```
    ].describe()
```

```
)
```

```
# Correlation
```

```
print("\nCorrelation:")
```

```
display(df.corr(numeric_only=True))
```

```
# Top 10 houses
```

```
print("\nTop 10 Houses by Median House Value:")
```

```
display(
```

```
    df.sort_values("median_house_value", ascending=False)
```

```
    .head(10)[
```

```
        [
```

```
            "longitude",
```

```
            "latitude",
```

```
            "median_income",
```

```
        "total_rooms",
        "median_house_value",
        "ocean_proximity"
    ]
]
)
```

Top 10 by income

```
print("\nTop 10 Records by Median Income:")
display(
    df.sort_values("median_income", ascending=False)
      .head(10)[
        [
            "longitude",
            "latitude",
            "median_income",
            "median_house_value",
            "ocean_proximity"
        ]
      ]
)
```

Export cleaned dataset

```
df.to_csv("housing_prices_cleaned.csv", index=False)
df.to_excel("housing_prices_cleaned.xlsx", index=False)
df.to_json("housing_prices_cleaned.json", orient="records", indent=4)

print("\nCSV, Excel and JSON files created successfully.")
```

OUTPUT

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   longitude              20640 non-null  float64
1   latitude               20640 non-null  float64
2   housing_median_age     20640 non-null  float64
3   total_rooms            20640 non-null  float64
4   total_bedrooms         20433 non-null  float64
5   population             20640 non-null  float64
6   households              20640 non-null  float64
7   median_income          20640 non-null  float64
8   median_house_value     20640 non-null  float64
9   ocean_proximity        20640 non-null  object
dtypes: float64(9), object(1)
memory usage: 1.6+ MB
```

Statistical description:

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income
count	20640.000000	20640.000000	20640.000000	20640.000000	20433.000000	20640.000000	20640.000000	20640.000000
mean	-119.569704	35.631861	28.639486	2635.763081	537.870553	1425.476744	499.539680	3.870671
std	2.003532	2.135952	12.585558	2181.615252	421.385070	1132.462122	382.329753	1.899822
min	-124.350000	32.540000	1.000000	2.000000	1.000000	3.000000	1.000000	0.499900
25%	-121.800000	33.930000	18.000000	1447.750000	296.000000	787.000000	280.000000	2.563400
50%	-118.490000	34.260000	29.000000	2127.000000	435.000000	1166.000000	409.000000	3.534800
75%	-118.010000	37.710000	37.000000	3148.000000	647.000000	1725.000000	605.000000	4.743250
max	-114.310000	41.950000	52.000000	39320.000000	6445.000000	35682.000000	6082.000000	15.000100

Missing values:

Ocean-Proximity-wise Housing Analysis:

	count	mean
ocean_proximity		
ISLAND	5	380440.000000
NEAR BAY	2290	259212.311790
NEAR OCEAN	2658	249433.977427
<1H OCEAN	9136	240084.285464
INLAND	6551	124805.392001

Housing Feature-wise Statistics:

	total_rooms	total_bedrooms	population	households	median_income	median_house_value
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	2635.763081	536.838857	1425.476744	499.539680	3.870671	206855.816909
std	2181.615252	419.391878	1132.462122	382.329753	1.899822	115395.615874
min	2.000000	1.000000	3.000000	1.000000	0.499900	14999.000000
25%	1447.750000	297.000000	787.000000	280.000000	2.563400	119600.000000
50%	2127.000000	435.000000	1166.000000	409.000000	3.534800	179700.000000
75%	3148.000000	643.250000	1725.000000	605.000000	4.743250	264725.000000
max	39320.000000	6445.000000	35682.000000	6082.000000	15.000100	500001.000000

Correlation:

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income
longitude	1.000000	-0.924664	-0.108197	0.044568	0.069120	0.099773	0.055310	-0.015176
latitude	-0.924664	1.000000	0.011173	-0.036100	-0.066484	-0.108785	-0.071035	-0.079809
housing_median_age	-0.108197	0.011173	1.000000	-0.361262	-0.319026	-0.296244	-0.302916	-0.119034
total_rooms	0.044568	-0.036100	-0.361262	1.000000	0.927058	0.857126	0.918484	0.198050
total_bedrooms	0.069120	-0.066484	-0.319026	0.927058	1.000000	0.873535	0.974366	-0.007617
population	0.099773	-0.108785	-0.296244	0.857126	0.873535	1.000000	0.907222	0.004834
households	0.055310	-0.071035	-0.302916	0.918484	0.974366	0.907222	1.000000	0.013033
median_income	-0.015176	-0.079809	-0.119034	0.198050	-0.007617	0.004834	0.013033	1.000000
median_house_value	-0.045967	-0.144160	0.105623	0.134153	0.049457	-0.024650	0.065843	0.688075

RESULT